Arista Access Node MAR-ACC-36 QoS Congestion Event Troubleshooting SOP

Node: MAR-ACC-36 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_7123

Description

This Standard Operating Procedure (SOP) outlines the troubleshooting and resolution steps for QoS Congestion Events on the Arista Access Node MAR-ACC-36 at the Marathahalli Core site. The failure domain in question is related to Microbursts, interface tail drops, and QoS queue starvation. This event is triggered by the ALARM_7123 alarm code, which indicates a Major severity failure. The underlying cause of this failure domain can be attributed to the inefficient allocation of QoS resources and the lack of prioritization of critical traffic flows on the Arista Access Node. The hardware mechanics of the failure can be attributed to the buffering capabilities of the Arista 7280SR-64C switch, which can lead to queue starvation and congestion when faced with high-velocity microbursts of traffic. Furthermore, the CLI dialect of the Arista switch can also contribute to the complexity of troubleshooting this issue, as the switch's command-line interface may not provide immediate visibility into the QoS configuration and resource allocation.

Pre-requisite Checks

- Check: Verify that the site location is Marathahalli Core and the node ID is MAR-ACC-36
- Check: Verify that the vendor CLI dialect is Arista and the role is Access
- Check: Verify that the alarm code is ALARM_7123 and the severity is Major
- Check: Verify that the redundant link status is active and ready for use
- Check: Verify that the remote hands are dispatched to the site location

Potential Root Causes

Cause	Probability
Inefficient allocation of QoS resources	High
Lack of prioritization of critical traffic flows	Medium
Buffering capabilities of the Arista 7280SR-64C switch	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos, Expected Output: QoS configuration and resource allocation, Description: This command will provide visibility into the QoS configuration and resource allocation on the Arista Access Node, helping to identify potential issues with QoS queue starvation and congestion
```

```
Command: show interface, Expected Output: Interface statistics and counters, Description: This command will provide detailed statistics and counters for each interface on the Arista Access Node, helping to identify potential issues with interface tail drops and congestion

Command: show mpls traffic-eng, Expected Output: MPLS TE configuration and statistics, Description: This command will provide visibility into the MPLS TE configuration and statistics on the Arista Access Node, helping to identify potential issues with critical traffic flows and prioritization
```

Resolution Steps

Step 1: Step: Reconfigure QoS policies, Command: configure terminal qos policy-map class target-queue exit exit , Description: Reconfigure the QoS policies to prioritize critical traffic flows and ensure efficient allocation of QoS resources

Step 2: Step: Adjust buffer sizes, Command: configure terminal interface buffer-size exit exit , Description: Adjust the buffer sizes on the Arista Access Node to reduce the likelihood of queue starvation and congestion

Step 3: Step: Verify MPLS TE configuration, Command: show mpls traffic-eng, Description: Verify that the MPLS TE configuration is correct and that critical traffic flows are being prioritized

Post-Check Verification

```
Command: show qos, Expected Output: QoS configuration and resource allocation, Description: Verify that the QoS configuration and resource allocation have been corrected

Command: show interface, Expected Output: Interface statistics and counters, Description: Verify that interface statistics and counters have improved

Command: show mpls traffic-eng, Expected Output: MPLS TE configuration and statistics, Description: Verify that the MPLS TE configuration and statistics have improved
```

Roll-back Procedures

- Step: Restore QoS policies, Command: configure terminal qos policy-map undo class undo target-queue exit exit , Description: Restore the QoS policies to their previous state

- Step: Restore buffer sizes, Command: configure terminal interface buffer-size undo exit exit , Description: Restore the buffer sizes to their previous state

- Step: Verify MPLS TE configuration, Command: show mpls traffic-eng, Description: Verify that the MPLS TE configuration has been restored to its previous state